Recycle Cardboard Into Anything With 3D Printing!

By XYZAidan in Workshop > 3D Printing 154,198 1,134 169 Featured Winner

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Paper is a ubiquitous and powerful material that we use every day. From newspaper to cardboard boxes to egg cartons, our world runs on paper, and **a lot** of it! However, all paper products tend to share a common characteristic: they're flat. Paper is so often 2D, flexible, and flimsy because of how it's manufactured. Paper is usually formed by suction, gravity, or rolling, but what if we used a different approach? What if we could **mold** paper, and compact it so that it has **volume**? We could make 3D objects that are **robust** and **reusable**, and we could make them from the tons of single-use paper thrown away every day!

In this guide, I'm going to show you how you can recycle paper and cardboard into almost any 3D object using very basic equipment: a 3D printer, a blender, and a vise.

YouTube Video:



Recycle Cardboard into Anything with 3D Printing!

XYZAidan





Watch on

If you feel so inclined, vote for this project in the Reuse Contest!

The process for molding paper into solid 3D forms is as follows:

- Blend the paper into a pulp with water
- Mix the pulp with a water-soluble binder
- Fill a 3D-printed mold with the pulp
- Use a pressing tool to compact the pulp with a vice
- Dry the molded pulp

This process of molding paper pulp can be used to create a variety of unique forms, dictated only by the geometry of the 3D-printed mold. With this guide, I am providing the designs for 7 molds:

- **60mm Coaster**
- **25mm Cube**
- **Triangle Surface**

- **Wave Surface**
- **Topographical Map of Mt. San Antonio**
- **Dish**
- **Desk Organizer**

The .STL files for these molds can be downloaded from Thingiverse.

This guide will show pictures from the process of multiple mold designs.

This project was inspired by an experiment by Will Haude of 3D Brooklyn. I've been toying with this project on-and-off for about two years now, working out the fine details and exploring the practical uses of this "material".

I believe that 3D-molded pulp has two primary benefits: it could be a replacement for plastic and it can made of end-cycle paper products that can no longer be recycled by traditional means.

Supplies

Equipment:

- 3D Printer
- Paper shredder (optional)
- Scissors/snips
- Gram scale
- Blender/food processor
- Cheesecloth (optional)
- Clamps
- Vise

Materials:

- Cardboard, newspaper, white paper, packing material, other paper (See Step #2)
- PVA glue, rice, or cornstarch (See Step #3)
- Water
- 3D printer filament (PLA works great)

Step 1: 3D Printed Molds



The first step to recycle paper into a new form is to create a mold. In this section, I'll explain the three parts of the molds, my own mold designs, and how you can design your own.

I used a system of three-part, 3D-printed molds to create most of my molded pulp. The three parts are:

Base

Wall - A straight-walled frame that is perpendicular to the base

Pressing Tool - A moving piece that is pushed down into the wall and compacts the paper pulp

My Molds:

All the .STLs for the mold pieces are available to download at the bottom of this step and on Thingiverse. Here is a guide for which files to download:

- 60mm Coaster (coasterMold1/2.STL)
- 25mm Cube (cube_base/cube_frame/cube_press.STL)

- Triangle Surface (Mesh_base/Mesh_frame/Mesh_press.STL)
- Sinusoidal Wave Surface (Box_base/Box_frame/Box_Sine_press.STL)
- Mt. San Antonio Topographic Map (Box_base/frame/Box_terrain_press.STL)
- Dish (Dish_Base/Dish_Wall/Dish_Press.STL)
- Desk Tray (Tray_Base/Tray_Wall/Tray_Press.STL)

For your first attempt at molding paper pulp, **I recommend you start with the 25mm cube.**

3D Print Parameters:

The mold parts need to withstand significant pressure. You may need to adjust your settings based on your printer, but here are the settings I used:

Base Piece Settings:

- 10-15% infill
- 2 shells
- 2-3 floor layers
- 3-4 ceiling layers

Wall Piece Settings:

- 15-20% infill
- 3 shells
- 2 floor layers
- 2 ceiling layers

Pressing Tool Settings:

- 10-15% infill
- 2 shells

- 2-3 floor layers
- 2-3 ceiling layers

All mold parts were printed in PLA filament.

Designing Your Own Molds:

Okay, now let's talk nitty-gritty details. Here are some features that I've found useful to include in my mold designs (**See picture**):

Wall-Press Tolerance

- The wall and the pressing tool need a slight gap between them to let out water but keep in pulp. For my printer, I found a tolerance of 0.4-0.5mm worked best.

Alignment Pegs

- Having pegs on the base piece that interface with holes on the wall piece keep the mold in tight alignment and strengthens the mold overall

Press Flange

- Having a lip or flange on the pressing tool sets a minimum height for the molded paper object and helps remove the pressing tool after molding

Fillets

- Molds will tend to crack at sharp edges and corners. Adding lots of fillets to your design will alleviate this.

Drainage

- Drain holes are only necessary on molds with large base surface area. I found drain holes with a diameter of 1-1.5mm work well.

Raised Base

- Having a raised section of the base that interfaces with the inside of the wall helps alignment and prevents the base from bowing under pressure.

Step 2: The Paper



Now you need paper to make pulp. Almost any type of paper works:

- Cardboard
- White paper
- Newspaper
- Food clamshells
- Formed packaging
- Egg cartons

The only paper you want to avoid is glossy paper and coated paper.

You can mix multiple types of paper together and even throw broken/failed paper moldings in with fresh paper.

Then, you have to make your paper into small blend-able pieces. A paper shredder works great for cardboard and sheets of paper. If you don't have a shredder, you can get away with using scissors. Cut the paper into >1in pieces.

Step 3: The Binder



Now you need to choose a binder material (glue) to help keep your molded paper pulp together. The binder improves the strength of the final object and prevents the material from coming apart as it dries. The binder needs to be water-soluble to mix with the pulp and saturate the paper fibers. I experimented with three types of binder: PVA glue, rice paste, and corn starch. Here are the pros and cons of each:

- **PVA glue**
  - Pros: It's the strongest binder, easy to obtain
  - Cons: Is plastic
- **Rice paste**
  - Pros: Organic
  - Cons: Takes time to make
- **Corn Starch**
  - Pros: Organic
  - Cons: Weakest binder, **must use boiling water when making the pulp**

For beginners, PVA glue is probably the easiest to use. My personal favorite is the rice paste because it is strong and organic.

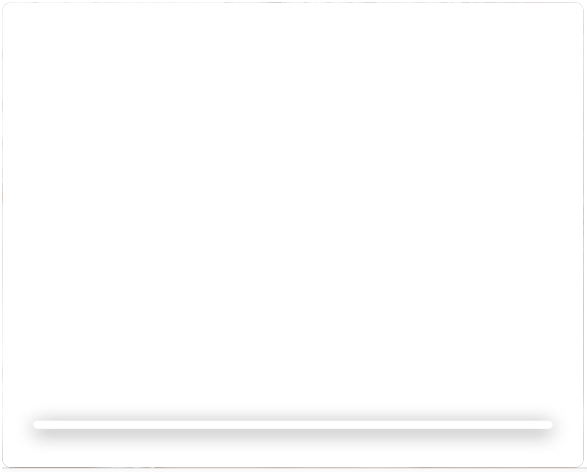
How to Make Rice Paste:

Take some pre-cooked white rice (short-grain or long-grain)

Simmer it with water and mash it with a whisk until its homogenous

Since you don't need a lot of paste to recycle paper, you can preserve extra paste by adding some clove oil and refrigerating.

Step 4: The Pulp



Now that you have the paper and the binder, it's time to turn it into pulp!

Mix the ingredients:

It's hard to predict how much paper you will need for a molding. The density of the paper fibers can vary depending on the mold geometry and your vice. Obviously, it's better to make too much pulp than not enough. Here's the average masses of my designs so you can estimate how much shredded paper to add to your pulp:

- Tray: 60g
- Disc: 10g
- Dish: 40g
- Wave: 35g
- Triangle: 25g
- Map: 25g
- Cube: Varies based on height

Ratios of paper to binder:

- PVA glue: 30g of paper : 25mL
- Rice paste: 30g of paper : 2 tablespoons
- Corn starch: 40g of paper : 2 tablespoons

Add your paper shreds and binder into a blender.

The last ingredient is the water. The goal here is to turn the paper and binder into a homogenous mixture **with as little water as possible**. Continually add small amounts of water into the blender and run the blender until it starts swirling and homogenizes.

Step 5: Pre-Pressing the Pulp



Before you can mold the pulp, you have to take some of water out first. If the pulp is too watery, it will mold poorly and will squeeze out of the mold (even explosively!).

You want to get the pulp to the consistency of wet clay. You don't need to get every drop of water out. I found wrapping the pulp in a cheesecloth and squeezing it worked great to press out the water. You can also work the pulp by hand to get water out.

Step 6: Molding the Pulp



Here we get to the real action! Assemble the mold base and wall. Reinforcing the mold with clamps will prevent bowing and distortion of the pulp, but may not be necessary depending on your vise and mold design.

Then, fill the mold with the pressed pulp you just made. Leave a bit of room at the top of the mold to help align the pressing tool. Push the pressing tool into the mold and make sure it is in line with the mold wall.

Put the whole assembly into the vise and crank it closed. You should tighten the vise **as far as it will go**. If you want to make the molded object thicker, then you can stop halfway-through pressing it and add more pulp.

Step 7: Drying



After about a day, you can remove the mold from the vise, remove the clamps, and pry off the pressing tool with a screwdriver. It will be suctioned to the pulp, so work around the edge and loosen it slowly. Once the pressing tool is out, leave the mold to dry for about a day.

You can then detach the base of mold. Let it dry some more. The molded paper pulp shrinks as it dries, so it will gradually loosen from the walls of the mold. Once it is dry to the touch and not soft, you should be able to push the molded paper object out. You can use the pressing tool to help push evenly.

You can put the object in front of a fan or vent to speed up the drying.

Step 8: Finishing



Once your molded paper pulp object has dried, it's time to pretty it up! Similar to any other molding process, molding paper pulp creates flashing, which is when the material creeps into the seems of the mold. You can trim this away with scissors (it cuts just like paper!).

The process sometimes creates rough faces on the object, but luckily the paper pulp sands easily with 60-grit sandpaper.

Lastly, if you need to make any holes or attach hardware to your molded part, the material can be drilled through easily.

Step 9: Results!



Look and Feel

The final molded pulp feels like paper on the surface, but it's volume gives it the feel of light wood (which makes sense given wood is just paper fibers with a binder of lignin). It's **extremely** rigid, and feels very comparable to plastic. It tends to keep the color of the paper it was made from, but does darken in spots that got a lot of pressure. Overall, the moldings look almost marbled, especially ones made from white paper/newspaper.

Strength

While I haven't done quantified strength tests on this material, I can anecdotally say that it is surprisingly strong. I can't break any of my moldings by hand. It took a number of hammer hits to significantly damage the desk organizer.

Shrinkage

As mentioned before, the molded paper deforms as it dries. The material expands in the axis of pressing and contracts in the other two axis. The amount of contraction depends on how densely the paper fibers are compacted, which will

depend on mold geometry. Pressing the mold harder and adding more pulp will reduce the amount of shrinkage.

Water Weakness

Given they're made of paper and water soluble glue, molded objects are obviously not water-resistant. Soaking it in water quickly causes it to break down.

Biodegradability

I would imagine the molded pulp would be compostable, given that paper and rice/corn starch are both compostable on their own. Testing needed.

Step 10: Variations and Improvements



I'm not done with this project yet! I've only really scratched the surface of what you can make by molding paper pulp. I hope that this Instructables teaches you enough to go out an experiment with recycling paper on your own and improve upon my work!

Anyways, here's some ideas I've had for new directions to explore:

- **Compostability**
  - Can someone chuck some of this stuff into a compost pile and see what happens? I see no reason why it wouldn't break down, as paper composts just fine (assuming the binder used is rice/corn starch/organic).
- **Solution to pre-pressing step**
  - The pre-pressing step is probably the biggest inefficiency in the process, as you have to remove water that already added. Maybe there's some alternative to a blender that could shred paper to pulp without as much water?
- **Waterproofing**

    - The biggest shortcoming of the molded paper compared to plastic is its weakness to water. I'd love to find some kind coating that could be applied to molded paper to make them resistant to water. Preferably the coating wouldn't be plastic or silicone and would be organic.

- **CNC machining**
    - If you molded large blocks out of paper, you could probably CNC mill the blocks and carve out details that you couldn't mold. Then, you could even recycle the chips and milling dust into new blocks!
- **Using an automatic press**
    - A hydraulic press instead of a vice could dramatically reduce the physical labor of the process and maybe even make the molded paper stronger.
- **Direct 3D printing**
    - It would no longer be molded pulp, but what if you could use paper pulp+binder in a paste extruder?

EDIT June 12th 2020:

Thank you all for all the support on this project. I'm truly honored to hear your interest in it. I want to keep a running list of people who have improved/innovated upon my recycling work:

- Chandler Hendry - Using large wooden molds to create cardboard panels
- Cask Tools - Creating custom packaging materials and using a vacuum chamber for molding
- Daniel Loo - Mold for organizational parts bins